

Disagree-to-Clarify: Surfacing Query Ambiguity through Structured Multi-Agent Disagreement

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Abstract

Conversational AI assistants routinely face ambiguous user requests, yet state-of-the-art LLMs typically presuppose a single interpretation rather than asking a clarifying question. Recent work addresses this by training models to recognize their own uncertainty, placing a strong introspective demand on the same model that produced the ambiguity-blind response. We propose Disagree-to-Clarify (D2C), a training-free multi-agent framework that surfaces ambiguity through structural disagreement instead. D2C instantiates three agents with distinct argumentative stances—a **Literalist**, a **Facet Finder**, and an **Intent Finder**—instructed to maintain and defend their readings across rounds. A synthesizer then either produces a targeted clarifying question or abstains when the agents converge. Our experiments on standard clarification benchmarks demonstrate that D2C matches the performance of state-of-the-art fine-tuned models, while exceeding them in LLM-judged quality. Furthermore, we show that disagreement topology is a robust diagnostic signal: distinct ambiguity types manifest as unique disagreement signatures’ across specialized agent axes, allowing the system to identify the nature of an information gap without explicit classification. Our code is available at: <https://github.com/jashparekh1/disagree-to-clarify>.

1 Introduction

Conversational AI assistants are routinely deployed in mixed-initiative settings where users articulate requests imperfectly. A user asking an assistant to “schedule a meeting with Alex” may have a specific Alex in mind that the assistant does not know about; a user asking for “healthy recipes” may mean low-calorie, low-sodium, or simply unprocessed. In each case the right move is not to commit to an interpretation, but to first clarify user intent. Asking clarifying questions has long been studied in conversational information seeking and task-oriented

dialogue (Aliannejadi et al., 2019, 2020; Rao and Daumé III, 2018), but the rise of strong instruction-tuned LLMs has shifted both the failure mode and the available tooling. State-of-the-art LLMs are biased toward immediate task completion: rather than acknowledging uncertainty, they presuppose a single interpretation, frustrating users who intended a different one (Zhang et al., 2024a; Chen et al., 2024). Existing solutions either train a single model to introspect on its own uncertainty (Andukuri et al., 2024; Zhang et al., 2024a; Chen et al., 2024), or apply multi-agent debate that asks agents to surface disagreement but urges the agents toward consensus (Du et al., 2024; Liang et al., 2024).

Our approach. We propose to invert the multi-agent paradigm. Disagree-to-Clarify (D2C) instantiates three agents with structurally distinct argumentative approaches—a Literalist who answers exactly what is asked, a Facet Finder who looks for aspects of the query that are under-determined, and an Intent Finder who reasons about the user’s unstated goals—and explicitly instructs them to maintain and defend their readings across multiple rounds of structured interaction. A synthesizer reads the resulting transcript and either generates a targeted clarifying question or abstains when the agents have converged. Figure 1 illustrates the full pipeline on a representative ambiguous query.

Hypothesis. We hypothesize that the specific “signature” of disagreement between different interpretive lenses provides a diagnostic signal for identifying the nature of a query’s ambiguity. Rather than relying on an external classification step, we propose that the *topology* of these internal conflicts—how and where competing perspectives diverge—reveals the underlying source of confusion. For instance, a tension centered on the query’s subject typically indicates a broad topic requiring subtopic specification, whereas a divergence regarding the user’s ultimate goal signals a missing

task-oriented element. In this framework, identifying the type of ambiguity becomes an emergent property of the dialogue structure: the specific “shape” of the disagreement identifies whether the gap in common ground resides in the query’s entities, the intended action, or the required context.

Contributions. This work makes two key contributions.

1. **Repurposing multi-agent architectures for ambiguity detection.** We invert the standard multi-agent debate objective: instead of treating disagreement as noise to be resolved, we treat it as the primary signal for whether a clarifying question is needed. By grounding the clarification logic in this structural tension, the system can bypass generic prompts and generate questions that are surgically targeted to the specific source of ambiguity.
2. **Characterizing ambiguity via disagreement topology.** We propose that the topology of inter-agent conflict provides a diagnostic signal for ambiguity types. We find that distinct information gaps manifest as observable “disagreement signatures” across specialized agent-pair axes, and show that the relative magnitude of structural tension between interpretive lenses can identify the structural nature of the underlying information gap.

2 Related Work

2.1 Clarifying Questions in Conversational AI

Clarifying questions have a long history in conversational information seeking and task-oriented dialogue (Purver et al., 2001; Rieser and Moore, 2005). Foundational benchmarks assess whether systems can identify ambiguous queries and request additional information: Qulac (Aliannejadi et al., 2019) and ClariQ (Aliannejadi et al., 2020) provide question-answer pairs grounded in TREC topics; ShARC (Saeidi et al., 2018) and Abg-CoQA (Guo et al., 2021) extend the setting to dialogue with rule-based or contextual clarifications. Rao and Daumé III (2018) and Zamani et al. (2020) study clarifying-question generation as a learning problem in its own right.

The LLM era has prompted new benchmarks tailored to model behaviour. CLAMBER (Zhang et al., 2024b) provides taxonomy-annotated ambiguous queries; ClarQ-LLM (Gan et al., 2026) introduces task-oriented bilingual dialogues in which

both seeker and provider are LLM agents, allowing the seeker’s clarification ability to be evaluated under diverse responses; HumanEvalComm (Wu and Fard, 2025) targets ambiguity in code-generation requirements; AGENT-CQ (Siro et al., 2024) uses LLMs to generate and evaluate clarifying questions for conversational search. ClarifyMT-Bench (Luo et al., 2025), contributes a multi-turn benchmark grounded in a five-dimensional ambiguity taxonomy (linguistic, intent, contextual, epistemic, interactional) specifically designed to exercise modern LLM’s ability to recognize and resolve ambiguity across dialogue history.

2.2 Teaching LLMs to Ask Clarifying Questions

Several recent methods train LLMs to ask better clarifying questions. GATE (Li et al., 2023) introduces generative active task elicitation, framing question-asking as preference acquisition. STaRGATE (Andukuri et al., 2024) fine-tunes a Questioner model on questions that increase the probability of high-quality oracle responses, yielding a 72% preference win-rate after two rounds of self-improvement; TO-GATE (Dou and Liu, 2026) extends this with trajectory optimization to suppress irrelevant questions. Zhang et al. (2024a) address a failure mode of preference labeling that conditions only on prior context: by simulating expected future turns, they reward clarifying questions that lead to interpretation-tailored answers, achieving a 5% F1 improvement on ambiguous open-domain QA. ACT (Chen et al., 2024) extends Direct Preference Optimization (Rafailov et al., 2023) with action-based contrast: dialogue actions—including the binary choice between clarifying and answering—are explicitly contrasted, improving disambiguation behavior.

Architecturally, this entire thread is single-agent: one model is trained to recognize its own uncertainty and produce a clarifying question. D2C departs from this lineage by spawning multiple agents with structurally distinct interpretive priors and treating their *disagreement*—rather than any individual agent’s introspective uncertainty signal—as the trigger for clarification. D2C is correspondingly training-free and model-agnostic: it can wrap any chat model, including small open models, with no preference data collection.

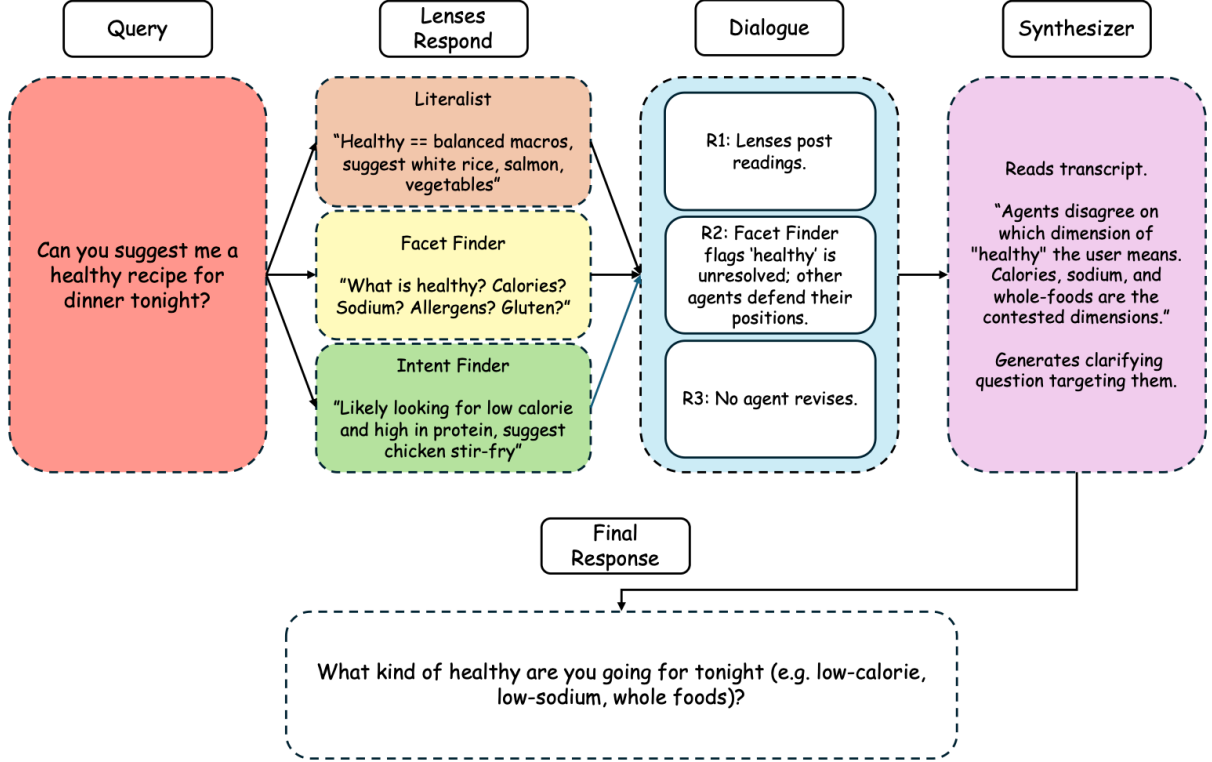


Figure 1: The Disagree-to-Clarify (D2C) pipeline. Given a potentially ambiguous user query, three specialized agents (Literalist, Facet Finder, Intent Finder) evaluate the prompt through forced adversarial lenses. The agents engage in a multi-round structured dialogue, holding, updating, or conceding their positions based on peer readings. Finally, a Synthesizer reads the resulting transcript and leverages the unresolved disagreement to generate a targeted clarifying question or outputs a CLEAR token if consensus is reached.

2.3 Multi-Agent LLM Frameworks

Multi-agent debate (MAD) (Du et al., 2024; Liang et al., 2024) has emerged as a prominent paradigm for improving LLM outputs on tasks with a single ground truth. Du et al. (2024) demonstrate gains on factual recall and arithmetic by having agents iteratively critique and revise each other’s answers; Liang et al. (2024) encourage divergent thinking via structured affirmative/negative debate adjudicated by a judge. Subsequent work has raised concerns about the paradigm: Yao et al. (2025) document sycophantic convergence in which agents abandon correct answers under peer pressure, and recent analyses suggest that many MAD frameworks do not consistently outperform single-agent baselines once compute is held constant.

A common feature of this literature is the assumption of a single ground truth—mathematical reasoning, factual QA, fact-checking. Under that assumption, consensus is the right objective and disagreement reflects error to be corrected. For ambiguous queries, where multiple interpretations may all be valid, this objective is misaligned: a suc-

cessful debate on an ambiguous question *discards* the very interpretive diversity the system needs to expose. D2C explicitly inverts the paradigm: agents are designed and prompted to maintain disagreement, and the disagreement structure itself—rather than a consensus answer—drives the system’s output. Persona- and role-based prompting has been used elsewhere to encourage diverse reasoning within a debate framework (Liang et al., 2024), but in those settings diversity is instrumental to reaching a better single answer; in D2C diversity is the answer.

3 Method

3.1 Problem Formulation

We frame clarification as a grounding move: an assistant cannot act faithfully on a request whose interpretation is not in common ground with the user (Clark and Brennan, 1991). Given a single, possibly ambiguous user turn u , the system must produce one of two outputs—a clarifying question q whose answer would supply at least some of the missing common ground, or a special token

CLEAR indicating no clarification is needed. This formulation is detection-aware: the same generation pathway handles both “ambiguous $\rightarrow$ ask” and “clear $\rightarrow$ proceed,” with the binary need-to-clarify decision recoverable from the output token rather than a separate classifier head.

3.2 D2C Architecture

Figure 1 summarizes the pipeline. D2C orchestrates three agents over N rounds of structured interaction, followed by a synthesizer that reads the resulting transcript and emits one move. We use $N = 3$ throughout.

Round 0. Each agent independently produces their interpretation of the ambiguous query. Agents do not see each other’s responses in this round.

Rounds 1 through $N-1$. Each active agent sees the previous round’s interpretations from all other active agents (conceded agents are listed as dropped, not shown) and emits a JSON object with three fields: a refined interpretation, a free-text `stance_reason`, and a stance from the set {HOLD, UPDATE, CONCEDE} (§3.4).

Synthesizer. After the final round, the synthesizer reads the full transcript together listing each remaining active agent’s stance and reason. Conceded agents are filtered out. The synthesizer outputs a JSON object containing either a clarifying question or the CLEAR token.

3.3 Agent Design: Forced Adversarial Stances

D2C uses forced adversarial stances: *each agent is assigned an argument it must defend*, creating structural tension across three primary axes.

LITERALIST (L) argues that the query is perfectly clear and has exactly one meaning, defending the dictionary-default reading.

FACET FINDER (F) argues that the query is a broad topic missing a subtopic or noun constraint (the “What”).

INTENT FINDER (I) argues that the query is missing an action or verb intent (the “Why/How”).

These lenses partition the space of potential confusion into mutually exclusive hypotheses. While our experimental data is sourced from the CLAMBER benchmark (Zhang et al., 2024b), we map its fine-grained categories into four primary *ambiguity dimensions*—*Linguistic, Intent, Facet*, and *Epistemic*—to evaluate how agent-pair divergence

(e.g., the L–F or F–I axis) serves as a diagnostic signature for each type of information gap.

3.4 Multi-Round Dialogue

To make the stance signal load-bearing, we constrain each agent’s revision behavior with an explicit decision tree, instantiated in the prompt. An agent emits HOLD (the default) if it can name a specific gap no other agent has captured; UPDATE if other agents have partially shifted its view but not fully covered its concern; and CONCEDE *only* if another agent’s reading covers its exact concern.

Two mechanisms protect against the agreement-cascade failure mode documented in plain multi-agent debate (Yao et al., 2025). First, concession is *sticky*: once an agent concedes, it drops out of all subsequent rounds and is not re-prompted. This prevents an agent from oscillating into a previously-conceded position under continued peer pressure. Second, the concession criterion is asymmetric per role—FACT FINDER concedes only if another agent proves its single reading is factually impossible, while INTENT FINDER concedes only if another agent identifies the same intent gap—which makes spurious convergence costly to produce. The dialogue terminates early if all remaining active agents concede in the same round, or if at most one active agent remains.

3.5 Synthesizer

The task of the synthesizer is only argument consolidation. Given the transcript and final-stance block, it decides which surviving argument best fits the query. If FACT FINDER survives with a convincing case, the synthesizer may emit CLEAR; if FACET FINDER or INTENT FINDER raises gaps the FACT FINDER could not refute, the clarifying question must target those gaps. This re-frames the synthesizer’s job as “which forced argument best fits this query” rather than “what gap do the three readings collectively imply,” aligning the decision criterion with the structure of the agent-level signal.

3.6 Implementation

Models and inference. The default agent and synthesizer backbone is Qwen-family chat models (Bai et al., 2023) served via Ollama (Marcondes et al., 2025) or a vLLM OpenAI-compatible endpoint (Kwon, 2025). Our reported numbers use qwen2.5:1.5b as the agent and synthesizer model and qwen3:4b as the LLM-as-a-Judge for evaluation, demonstrating that the pipeline runs reliably

Method	F1	Quality $\uparrow$	Divergence $\uparrow$	Similarity $\uparrow$	Coverage%	Rounds $\downarrow$
vanilla	0.35	1.8	0.00	0.246	23.1	0.0
self_consistency	0.34	1.8	0.00	0.244	22.8	0.0
sft	0.79	3.7	0.00	0.663	79.9	0.0
rl	0.79	3.7	0.00	0.668	81.1	0.0
parallel	0.64	2.7	0.28	0.387	51.5	1.0
d2c (ours)	0.79	3.9	0.29	0.690	91.6	2.1

Table 1: Overall performance of the Disagree-to-Clarify (D2C) framework compared to zero-shot, parallel, and fine-tuned baselines. The table reports ambiguity detection F1, LLM-as-a-Judge Quality, Internal Divergence, Semantic Similarity to gold-standard questions, Coverage percentage, and Average Debate Rounds. D2C achieves performance on par with fine-tuned models (SFT, RL) across detection metrics while yielding the highest question quality and semantic similarity.

on small open-weight models.

Sampling. Agents use temperature 0.7 to encourage genuine interpretive variation across the three roles; the synthesizer uses 0.3 for more deterministic adjudication; the LLM judge uses 0.0.

4 Experiments

4.1 Baselines

We compare D2C to several baselines on question-asking benchmarks to understand its performance. The models evaluated, together with the references to the prompts, are:

1. **Vanilla:** A standard zero-shot baseline where an agent outputs a clarifying question for ambiguous queries or CLEAR for unambiguous ones (A.1).
2. **Self-Consistency:** Runs the vanilla approach three separate times on the same query at a relatively high temperature ($T = 0.7$), and keeps the response whose embedding is closest to on average to the other runs. This removes the most extreme responses.
3. **Parallel Baseline:** Both the *Parallel* baseline and D2C share the same three specialized roles. The difference is the interaction: Parallel runs agents in isolation for one round, while D2C allows agents to see each other’s outputs and debate for up to three rounds. (A.2).
4. **SFT:** An imitation learning baseline where the 1.5B model is fine-tuned via LoRA on expert-level clarifying questions. These targets are generated through knowledge distillation from a 8B teacher model (llama3.1:8b), which was tasked with synthesizing disjunctive questions based on multiple ground-truth

interpretations from the AmbigQA training set (A.3).

5. **RL:** A preference-tuned baseline using Direct Preference Optimization (DPO). The model is fine-tuned on a preference dataset where candidate clarifying questions are ranked by a “Resolution Judge” reward signal. This signal scores questions based on their simulated effectiveness in resolving ambiguity during a future conversation turn (A.4).

4.2 Datasets

Our approach is evaluated on the following datasets:

1. **QULAC:** One of the first datasets to focus on asking clarifying questions to improve results in a web search domain (Aliannejadi et al., 2019). It consists of over 10K broad queries (e.g. “becoming a paralegal”) together with several gold clarifying questions that disambiguate them.
2. **ClariQ:** Extended previous datasets by introducing a metric indicating *when* to ask a question and not only *what* to ask (Aliannejadi et al., 2020). This is done through a *clarification need* label ranging from 1 to 4.
3. **Abg-CoQA:** Extended the task of asking clarifying questions to a conversational domain (Guo et al., 2021). In it, the ambiguity can arise over different turns and the clarifying questions are specific to them.
4. **CLAMBER:** Clarification benchmark focused on LLM conversations. It introduced a taxonomy designed to understand misalignment between the models knowledge and the user prompts (Zhang et al., 2024b).

Variant	F1	Quality $\uparrow$	Divergence $\uparrow$	Similarity $\uparrow$	Coverage% $\uparrow$
full_d2c	0.73	3.91	0.28	0.678	91.3
- no_fact_finder	0.73	3.74	0.37	0.671	83.7
- no_facet_finder	0.55	2.60	0.20	0.413	50.0
- no_intent_finder	0.68	3.62	0.23	0.596	81.5
generic_agents	0.44	2.07	0.04	0.297	32.6
collaborative	0.49	2.24	0.11	0.328	37.5

Table 2: Impact of individual agent roles and interaction protocols on D2C clarification performance ($N = 320$). The top block isolates the contribution of each agent role; removing the Facet-Finder causes the largest drop across all metrics. The bottom block confirms that both role specialisation and forced adversarial stances are necessary: replacing specialised roles with generic agents (*generic_agents*) or softening prompts to allow free concession (*collaborative*) substantially degrades performance across all metrics.

4.3 Evaluation Metrics

- **Detection F1:** Ambiguity detection F1.
- **Quality (LLM-as-Judge Quality):** A subjective quality score on a scale of 1–5, assigned by a "Judge" model (e.g., Qwen3:4b). It evaluates how helpful, natural, and relevant the clarifying question is relative to the user’s original query and a gold-standard reference (A.6).
- **Divergence (Internal Divergence):** A semantic measure of the residual disagreement between agents. Calculated as the average pairwise distance ($1 - \text{cosine similarity}$) between agent interpretations in the final round of deliberation. A high Div. score indicates that multiple distinct interpretations persisted despite the dialogue process, serving as a quantitative proxy for the "intensity" or "hardness" of the ambiguity being surfaced to the synthesizer.
- **Similarity (Semantic Similarity):** The maximum cosine similarity between the sentence embeddings of the generated question and the set of human-authored gold-standard questions. It ranges from 0 to 1, where higher values indicate the model is capturing at least one valid semantic interpretation found by human annotators.
- **Coverage (Interpretation Coverage):** A binary assessment (0 or 1) assigned by the Judge model, indicating whether the generated question successfully encompasses and allows the user to distinguish between *all* primary interpretations identified in the gold reference. The

values reported in our results represent the percentage of successful assessments across the evaluation set. covers multiple possible interpretations (facets) of an ambiguous query.

- **Rounds (Average Dialogue Rounds):** The average number of internal "agent turns" or dialogue cycles executed before the model synthesized its final output:
 - *Vanilla* and *Parallel* uses 0 rounds (direct generation).
 - *D2C* utilizes dynamic multi-round deliberation (averaging 2.8 rounds in this run).

5 Results

Table 1 demonstrates that D2C significantly outperforms zero-shot and single-agent baselines, achieving performance highly competitive with heavily fine-tuned models. Specifically, D2C achieves an F1 score of 0.79 and a Quality score of 3.9, compared to the Vanilla baseline’s F1 of 0.35 and Quality of 1.8. Strikingly, the training-free D2C framework matches the F1 score (0.79) of both the Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL) baselines while exceeding them in LLM-as-a-Judge Quality (3.9 vs. 3.7) and Coverage (91.6% vs. 81.1%).

The results also validate the necessity of multi-round deliberation. The Parallel baseline, which utilizes the same three specialized roles but prevents inter-agent communication, achieves an F1 of only 0.64 and a Quality of 2.7. By allowing agents to debate for an average of 2.1 rounds, D2C increases the F1 score by 15 basis points, confirming that the topology of the dialogue itself is crucial for extracting the correct ambiguity signal.

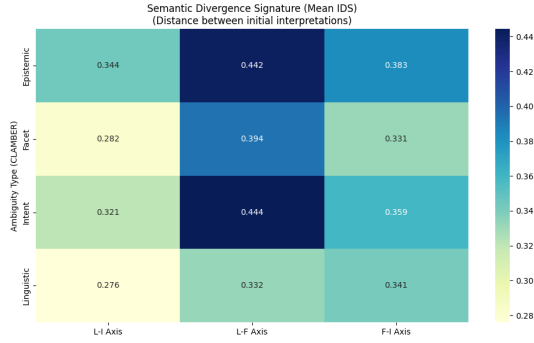


Figure 2: Relationship between disagreement topology and ambiguity type

The `self_consistency` baseline (F1 = 0.34, Quality = 1.8, Coverage = 22.8%) performs nearly identically to the vanilla single-call baseline on all metric. This is a notable negative result: generating multiple samples at higher temperature and selecting the centroid response provides no meaningful benefit over a single call. Repeated sampling converges to the same interpretation rather than surfacing competing readings, confirming that stochastic diversity alone is insufficient to detect ambiguity and structured disagreement is necessary.

5.1 Ablations

5.1.1 Impact of role and interaction on performance.

To understand the contribution of individual components, we evaluate several architectural ablations in Table 2. The results reveal an asymmetry across the three interpretive lenses. Removing the Fact-Finder has a negligible effect (F1: 0.73, Quality: 3.74), suggesting the “query is clear” anchor is redundant, though the notably higher divergence score (0.37) indicates the remaining two agents maintain stronger disagreement without it. By contrast, removing the Facet-Finder causes the steepest degradation across all metrics (F1: 0.55, Quality: 2.60, Coverage: 50.0%), confirming that subtopic disambiguation is the primary driver of question quality. Removing the Intent-Finder produces a moderate but consistent drop (F1: 0.68, Quality: 3.62), indicating that action-intent disambiguation is a secondary but meaningful signal.

We further validate our forced adversarial stance design against two alternative interaction protocols. Replacing specialised roles with generic, unconstrained agents collapses interpretive divergence to near zero (Div: 0.04), yielding the worst performance across all metrics (F1: 0.44, Quality:

2.07, Coverage: 32.6%), confirming that without role specialisation agents converge to similar default readings. Instructing agents to seek consensus (collaborative) yields notably higher divergence (0.11) and slightly better F1 (0.49) than generic agents, yet remains far below the full system, demonstrating that role specialisation and forced adversarial stances jointly drive performance, and that softening either one substantially degrades coverage and question quality.

5.1.2 Characterizing Disagreement Signatures

We investigate whether the topology of agent disagreement provides a diagnostic signal for identifying the underlying nature of the ambiguity. Figure 2 visualizes the *Semantic Divergence signature*, which we quantify via the *Mean Interpretive Disagreement Score* (IDS).

Let $i_r(q)$ denote the initial interpretation of a query q by an agent with role $r \in \{L, F, I\}$ (Literalist, Facet, Intent). We define the semantic distance between any two agents as $d(i_{r_1}, i_{r_2}) = 1 - \cos(\phi(i_{r_1}), \phi(i_{r_2}))$, where $\phi(\cdot)$ is a sentence embedding function and $\cos(\cdot, \cdot)$ is the cosine similarity. For each ambiguity dimension D and agent-pair axis $a = (r_1, r_2)$, the Mean IDS is defined as the expected divergence over all queries in that category:

$$\text{Mean IDS}(D, a) = \frac{1}{|D|} \sum_{q \in D} d(i_{r_1}, i_{r_2}) \quad (1)$$

The results demonstrate that distinct information gaps manifest as characteristic patterns of structural tension. For instance, while most categories show a high degree of conflict on the $L-F$ axis, *Linguistic* ambiguities (e.g., polysemy or co-reference) exhibit a unique relative peak on the $F-I$ axis. This suggests that the relative magnitude of conflict between specific agent-pair axes provides a “fingerprint” of the query’s structural nature, allowing the system to distinguish between surface-level referential gaps and deeper goal-oriented omissions without an explicit classification step.

6 Conclusion

In this work, we introduced Disagree-to-Clarify (D2C), a novel multi-agent framework that surfaces ambiguous user queries through structured disagreement rather than introspective uncertainty. By assigning agents forced adversarial stances—a LITERALIST, a FACET FINDER, and an INTENT

FINDER—we ensure the preservation of interpretive diversity across multi-turn dialogue. Our evaluation on standard clarification benchmarks demonstrates that D2C significantly outperforms zero-shot single-agent models and achieves performance comparable to state-of-the-art fine-tuned models, reaching an F1 score of 0.79 without any task-specific training. Furthermore, our ablation studies validate that both the rigid argumentative stances and the multi-round interaction protocol are essential to this performance; removing specialized lenses or encouraging collaborative consensus universally degrades the system’s ability to detect and resolve ambiguity. We also show that the *topology* of the resulting disagreement serves as a diagnostic signal, where the specific “shape” of inter-agent conflict characterizes the structural nature of the underlying information gap. Ultimately, D2C demonstrates that for tasks lacking a single ground truth, the topology of multi-agent disagreement is itself a powerful, structured signal.

7 Contribution Statement

Jash Parekh developed the D2C system and contributed to the writing of the paper. Javier Diego contributed to the design of the system, ran experiments, and contributed to the writing of the paper. Akash Elumalai ran extensive experiments, contributed to development of D2C code and optimized the system to run on vLLM.

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A Prompt Templates

This section details the core instructions and architectural setups for each method in the final evaluation.

A.1 Vanilla Baseline (Single-Turn)

Vanilla CQG Prompt

```
Analyze the following query: "{query}"

If it needs clarification to be answered accurately, generate ONE concise clarifying question.
If it is clear and requires no further information, output: "CLEAR"

Output ONLY this JSON:
{"clarifying_question": "<your question or CLEAR>"}
```

A.2 Parallel and D2C (Multi-Agent Interaction)

Fact-Finder (Role A)

You are the FACT-FINDER. You MUST argue that the user's query is perfectly clear and has only ONE meaning. Explain what that meaning is and why no clarification is needed. 1-2 sentences only. Even if you suspect ambiguity, you must defend the position that it is clear.

Facet-Finder (Role B)

You are the FACET-FINDER. You MUST argue that this query is a broad topic and the user is missing a 'Subtopic' or 'Facet' constraint. Identify what those missing facets are. 1-2 sentences only. You must defend the position that the query is too broad.

Intent-Finder (Role C)

You are the INTENT-FINDER. You MUST argue that this query is missing an 'Action' intent (e.g., buying vs. learning, searching vs. creating). Identify what the missing actions are. 1-2 sentences only. You must defend the position that the intent is missing.

A.3 SFT

This is the template used to reframe the gold clarifying question to generate the SFT dataset.

SFT reframe template prompt

You are an expert at creating clarifying questions for ambiguous queries.
Given an original query and several disambiguated interpretations, create a single, concise, and natural clarifying question that would help a user specify which of these interpretations they meant.

The question should:

1. Be polite and natural.
2. Explicitly mention the different interpretations to help the user choose (e.g., "Are you asking about X or Y?").
3. Be concise and avoid conversational filler.

Original Query: {query}

Ground Truth Interpretations:
{interpretations}

Clarifying Question: ""

A.4 Fine-tuned Baselines (SFT and RL)

Unlike the multi-agent approaches, the SFT and RL models are end-to-end generators. Their "policy" is baked into the model weights through training. During evaluation, they use the minimal ChatML format to trigger the behavior learned during fine-tuning.

Fine-tuned Minimal Prompt

```
<|im_start|>user  
{query}<|im_end|>  
<|im_start|>assistant
```

Note: The SFT model was trained on expert-synthesized disjunctive questions distilled from a 7B teacher model. The RL (DPO) model was trained on preferences generated by the Resolution Judge reward signal.

A.5 Synthesizer (Decision Logic)

Both the Parallel and D2C methods use this system to consolidate the multi-lens interpretations into a final output.

D2C Synthesizer Prompt

You read three agents' forced arguments about a user query. Your goal is to resolve the conflict by presenting the divergent arguments as a choice to the user.

1. Use a disjunctive "Are you asking about [Facet A] or [Action B]?" structure.
2. BE CONCISE: Under 20 words. No preamble.
3. VALIDATION: If the FACT-FINDER proved the query is clear, output ONLY "CLEAR".

A.6 LLM-as-a-Judge Evaluation

Judge System Prompt

You are an expert evaluator of clarifying questions for ambiguous user queries. Your task: Evaluate a clarifying question generated by a system given an ambiguous query and its known possible interpretations (disambiguated versions).

The clarifying question should:

1. Be concise and natural.
2. Directly address the core ambiguity that separates the provided interpretations.
3. Help a user quickly identify which of the interpretations they meant.

Provide your evaluation in JSON format:

```
{  
  "score": (1-5 scale),  
  "reasoning": "A brief explanation of why  
    this score was given.",  
  "covers_interpretations": (true/false,  
    whether the question allows  
    distinguishing between ALL provided  
    interpretations)  
}
```

Judge User Prompt

Original Ambiguous Query: {query}

Possible Interpretations:
{interpretations}

Generated Clarifying Question:
{clarifying_question}

Evaluate the generated question based on how well it helps resolve the ambiguity between the provided interpretations.

A.7 Ablation Study Prompts

The ablation study tests six variants of the D2C pipeline. Below are the prompts that differ from the full system.

A.7.1 Role-Removal Variants

(no_fact_finder, no_facet_finder, no_intent_finder)

These variants run the dialogue with only two of the three D2C agents (the named role is dropped). Because the synthesizer assumes three-agent transcripts, synthesis is performed directly with a role-agnostic synthesizer:

Role-Agnostic Synthesizer (2-Agent Variants)

You read multiple agents' interpretations of a user query.
If the agents disagree significantly, formulate one clarifying question that resolves the key point of disagreement. Under 20 words. No preamble.
If the agents agree the query is clear, output ONLY the word CLEAR.
Output only the question or CLEAR.

A.7.2 Generic Agent Variant

(generic_agents)

All three agents use an identical, taxonomy-free prompt with no role specialisation. This isolates whether the Fact/Facet/Intent taxonomy contributes beyond generic ambiguity analysis. The forced-stance instruction below replaces the default concession policy so that agents default to HOLD:

Generic Agent Prompt (all three agents)

You are an AI assistant. Analyze the user's query and identify any potential ambiguities. If it is clear, say CLEAR. If not, explain one specific interpretation. 1-2 sentences only.

Forced Stance Instructions (generic_agents & collaborative variants)

Decide your stance:
1. Has another agent explicitly and EXACTLY addressed your specific argument (not just a related topic)? -> CONCEDE only then.
2. Otherwise -> HOLD. Do not concede because of surface overlap or partial agreement.

A.7.3 Collaborative Variant (collaborative)

Agents retain the Fact/Facet/Intent role names but use permissive prompts that allow free concession. Combined with the forced-stance instructions above (same scaffold as generic_agents), the only difference from generic_agents is that agents are given role-specific lenses. This tests whether adversarial pressure in the system prompt matters independently of the role taxonomy:

Collaborative Agent Prompts — Fact-Finder

Your goal is to see if the query is clear. If you find real ambiguity, you may admit it, but your default lens is to look for a clear reading. 1-2 sentences only.

Collaborative Agent Prompts — Facet-Finder

Your goal is to see if subtopics are missing. If the query is already specific, you may admit it is clear. 1-2 sentences only.

Collaborative Agent Prompts — Intent-Finder

Your goal is to see if the action is missing. If the intent is obvious, you may admit it is clear. 1-2 sentences only.